SyllaByte: AI-Powered Micro-Learning Video Generation from Course Syllabi and Lectures

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ABSTRACT

University students increasingly struggle to engage with dense lecture recordings and lengthy syllabi. While micro-learning and AI-generated educational video have each been studied in isolation, no system has yet automatically generated short, syllabus-aligned, personalized micro-videos directly from a professor's lecture content and course outline. This proposal introduces SyllaByte, a multi-agent AI pipeline that ingests a course syllabus and lecture recordings, segments them into knowledge units, and generates 2–5 minute animated micro-videos calibrated to each student's assessed prior knowledge. The system employs four specialized components: a Curriculum Alignment Agent that maps lecture content to Bloom's Taxonomy levels, a Student Knowledge Assessment Agent using Computerized Adaptive Testing, a Script Generation Agent grounded in Cognitive Load Theory, and an Animation and Video Synthesis Agent producing Manim-based visualizations for STEM content. We ground the proposal in fourteen reviewed papers spanning multi-agent video generation, micro-learning design, adaptive personalization, and instructional methodology. We propose two primary research questions, a four-component system architecture focused on STEM content, and a rigorous mixed-methods evaluation plan including a pre-registered semester-long pilot study with automatic mid-semester student profile reassessment.

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KEYWORDS

micro-learning, AI-generated video, personalized learning, Bloom's taxonomy, cognitive load theory, multi-agent systems, Manim, computerized adaptive testing, STEM education, knowledge unit graph

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1 INTRODUCTION

Student disengagement with lecture recordings is one of the most persistent and well-documented challenges in higher education. Research in cognitive psychology consistently shows that attention and working memory degrade sharply after 10–15 minutes of passive instruction [11], yet most university lectures run 60–90 minutes and are uploaded to learning management systems as monolithic, unadapted recordings. Students are left to manage their own pacing, depth, and comprehension without any structural support calibrated to their individual knowledge level.

This problem has grown more acute in the digital era. The widespread adoption of short-form content platforms such as YouTube Shorts, Instagram Reels, and TikTok has fundamentally adjusted student expectations for content density and pacing [4]. Students accustomed to highly engaging, bite-sized content find it increasingly difficult to sustain attention through hour-long recordings delivered without interaction or adaptation. The result is shallow learning, poor long-term retention, and growing disengagement from course material — a trend that has negative consequences for academic performance.

Why This Research Matters. Micro-learning — the delivery of educational content in short, focused segments — has emerged as a direct response to this challenge. Empirical evidence strongly supports its effectiveness: the Adaptive Microlearning study [5] found that structured 2–6 minute micro-videos increased immediate retention by 15 percentage points (85% vs. 70%) and one-week delayed recall by

25 points (80% vs. 55%) compared to traditional full-length video. ReelEd [10] demonstrated that LLM-generated short-form videos achieved quiz scores of 93.85% versus 79.72% for long-form video, with quiz completion time 26% faster. These findings provide compelling motivation for investigating automated micro-video generation as a scalable solution to the engagement crisis in higher education.

Despite this promise, the existing literature reveals a critical and unresolved gap. Current AI-generated video systems such as LASEV [14] and Code2Video [3] achieve impressive results for K-12 mathematics and STEM but generate videos from generic problem statements — not from a professor's own lecture content. ReelEd [10] clips existing YouTube videos rather than creating new content tailored to a specific course. The Adaptive Syllabus paper [13] generates structured syllabi with AI but does not produce videos at all. The AI Educational Video Methodology paper [1] provides a practitioner pipeline for university courses but relies entirely on manual scripting with no automation. Critically, no existing system automatically generates short-form videos grounded in a professor's own syllabus and lecture recordings. SyllaByte directly targets this gap.

Background. Three well-established theoretical frameworks collectively motivate the design of SyllaByte. Cognitive Load Theory (CLT) [11] states that human working memory is limited and that instructional design must minimize extraneous cognitive load while maximizing germane load — the load that supports schema formation. The CLT-Based Lecture Model (CLT-bLM) [5] operationalizes CLT into a four-phase instructional cycle: Prepare (activate prior knowledge), Initiate (state learning objectives and provide analogies), Deliver (present essential content in manageable chunks), and End (recap and prompt reflection). Mayer's Multimedia Learning Principles [6] further establish that people learn better from coordinated words and visuals than from words alone, directly motivating the use of animated video over text-only formats. Finally, Bloom's Taxonomy [2] provides a hierarchical framework for calibrating learning objectives — from basic recall to higher-order analysis and creation — enabling content to be precisely matched to each student's assessed knowledge level.

Challenges. Building SyllaByte requires overcoming four major technical and pedagogical challenges. First, *automatic curriculum parsing* is technically challenging: extracting structured, pedagogically coherent knowledge units from unstructured lecture audio and PDF syllabi requires robust speech recognition, layout parsing, and semantic segmentation. Second, *reliable animated video generation from code* is still an unsolved challenge: the LLM+Manim study [9] reported a 38% render failure rate when using large language models to generate Manim animation code, and hallucinated mathematical symbols are a known failure mode that is fatal for STEM educational content. Third, *meaningful personalization* is difficult to achieve in practice: most existing systems assign students a single initial knowledge level and never update it, leading to personalization drift as students' understanding evolves throughout the semester. Fourth, *rigorous longitudinal evaluation* of AI-generated educational video has not yet been conducted: No paper reviewed in this literature reports controlled studies that track learning outcomes over a full semester [3, 5, 10, 14]. SyllaByte is designed to directly address each of these challenges through its four-component multi-agent architecture and its automatic mid-semester student reassessment mechanism.

2 RELATED WORK

Prior work on AI-driven educational video generation and micro-learning falls into five thematic clusters, each of which SyllaByte directly addresses.

2.1 Multi-Agent AI Video Generation Systems

Recent work has demonstrated the value of decomposing video generation into specialized agents rather than relying on monolithic models. LASEV [14] introduced an LLM-based multi-agent system for educational video generation that distributes reasoning, scripting, illustration, and critique across dedicated agents, employing a three-critique loop (semantic, tool, and rule) to validate outputs at each stage. The system achieved over 95% cost reduction compared to industry-standard pipelines while maintaining production quality. Code2Video [3] extended this approach with a code-centric multi-agent framework, introducing the ScopeRefine hierarchical debugging strategy that progressively repairs Manim rendering errors at the line, block, and section levels. Code2Video also proposed the TeachQuiz benchmark, in which a vision-language model answers questions before and after watching generated videos to provide an automated proxy for learning efficacy. Both systems, however, generate videos from generic problem statements rather than from a professor's actual lecture content. SyllaByte addresses this by grounding the content on a professor's content.

2.2 Micro-Learning Design Frameworks

The authors operationalize Cognitive Load Theory through the CLT-bLM four-phase instructional cycle (Prepare, Initiate, Deliver, End), tag content at Bloom's Taxonomy levels, and use Computerized Adaptive Testing with Item Response Theory for student profiling. Their evaluation showed that structured 2–6 minute micro-videos increased immediate retention by 15 percentage points and one-week delayed recall by 25 points compared to traditional lengthy video, with a System Usability Scale score of 82.1. However, the

study had no longitudinal evaluation, was limited to a single institution and discipline, and required manual curation. The AI-Driven Personalized Microlearning Framework [5] (Wiley CAE, 2025) introduced a merged hybrid prompting strategy combining instruction tuning, step-by-step reasoning, prompt chaining, and persona pattern, and demonstrated a Decision Table ML model achieving high accuracy on student outcome prediction using the OULAD dataset. SyllaByte adopts both the CLT-bLM structuring and the merged hybrid prompting strategy while extending personalization to per-Knowledge-Unit granularity rather than a single global learner level.

2.3 Adaptive Syllabi and Personalization

The Adaptive Syllabus Generation paper [13] demonstrated that AI can parse course materials and generate structured, phase-annotated syllabi with per-session learning objectives. While it is valuable, this system does not produce videos at all, leaving a disconnect between syllabus structure and content delivery. SyllaByte's Curriculum Alignment Agent directly bridges this gap by mapping each Knowledge Unit extracted from the syllabus to a Bloom's Taxonomy level, cognitive load estimate, and target video duration. ReelEd [10] demonstrated that LLM-curated short-form educational videos achieved quiz scores of 93.85% versus 79.72% for long-form video and developed the trust-in-automation scale used as a secondary outcome measure in our evaluation. However, ReelEd clips and remixes existing YouTube videos instead of generating new content. This means that it cannot adapt to the syllabus of a specific course or to the instructional style of a professor.

2.4 Code-Based Animation and STEM Video Synthesis

The LLM+Manim study [9] systematically evaluated LLMs for generating Manim animation code and documented a 38% render failure rate even with state-of-the-art models. The authors introduced ManimBench, a curated dataset for fine-tuning compact models, and demonstrated that LoRA fine-tuned 7B-parameter models could achieve 61% render success competitive with much larger base models. The study also identified API hallucinations as the primary failure mode and noted that no visual or pedagogical quality evaluation was performed. SyllaByte directly responds to these limitations by adopting a constrained Manim API subset, integrating ScopeRefine-based debugging, and adding a vision-language Critic agent for quality verification. SciTalk [13] [7] further demonstrated that creator-inspired iterative feedback loops outperform single-pass generation for scientific short-form video. This validates SyllaByte's choice of a multi-agent feedback architecture.

2.5 Instructional Methodology and Pedagogical Frameworks

The AI Educational Video Methodology paper [1] provides a educator-oriented pipeline for university video creation grounded in Mayer's Multimedia Learning Principles, including specific tool recommendations (ElevenLabs and Azure TTS for narration) and a human-in-the-loop quality gate. However, the pipeline relies entirely on manual scripting and tool chaining with no automation. The Generative AI Pedagogical Framework [1] [12] proposed xAPI behavioral analytics for tracking student engagement and a Bloom's-aligned interactivity module, while flagging personalization drift as a significant limitation in current adaptive systems. The AI-Generated Instructional Videos Systematic Review [1] [8] synthesized findings across studies and identified social presence deficit as a recurring concern — AI-generated videos consistently score lower than human-instructor videos on perceived social presence, motivating SyllaByte's optional professor voice-over feature.

2.6 Synthesis and Gap Analysis

Four persistent limitations emerge across these clusters: no system ingests a professor's own course materials as primary input; personalization is typically static with no automated update mechanism; STEM-focused systems suffer from high Manim render failure rates; and no study has conducted a controlled semester-long evaluation. SyllaByte's four-component architecture is designed to address each of these gaps directly.

3 METHODOLOGY

SyllaByte is a four-component multi-agent pipeline focused on STEM content. The multi-agent design is grounded in evidence from LASEV [14], Code2Video [3], and the broader literature showing that distributing reasoning, scripting, and critique across specialized agents consistently outperforms single-agent generation. Each component directly addresses a specific wekaness identified in the reviewed literature.

3.1 Component 1 — Ingestion and Curriculum Alignment Agent

The pipeline begins by ingesting two document types: a structured course syllabus in PDF or Word format, and lecture recordings in video or audio form. The Ingestion Agent uses OpenAI Whisper [5] for automatic speech recognition of lecture audio, extracts slide content from any embedded presentation using a layout parser, and segments the full course into a Knowledge Unit Graph (KUG) — a directed acyclic graph where each node is a discrete teachable concept and edges encode prerequisite relationships.

The Curriculum Alignment Agent then maps each Knowledge Unit to four attributes: (a) its position in the syllabus outline, (b) a Bloom's Taxonomy level [2], (c) an estimated cognitive load rating grounded in Cognitive Load Theory [11], and (d) a recommended target video duration of 2–5 minutes. This agent extends the Adaptive Syllabus paper [13] by linking syllabus structure directly to video generation.

3.2 Component 2 — Student Knowledge Assessment Agent

Before any video is delivered, the system administers a Computerized Adaptive Test (CAT) modeled after the Item Response Theory (IRT) approach validated in the Adaptive Microlearning paper [5]. The assessment dynamically adjusts difficulty based on student responses, converging on a reliable estimate of prior knowledge for each Knowledge Unit in fewer questions than a fixed-form test. Student ability θ is estimated using the 2-Parameter Logistic IRT model:

$$P(\text{correct}) = \frac{1}{1 + e^{-\alpha(\theta - \beta)}}$$

where α is the discrimination parameter and β is item difficulty. The resulting student profile assigns each KU a readiness level (novice, developing, proficient) and maps it to the appropriate Bloom's Taxonomy tier — novice maps to Remember/Understand, developing to Apply/Analyze, and proficient to Evaluate/Create.

Automatic mid-semester reassessment. At the mid-semester point (Week 7), the system automatically re-administers the assessment using a combination of a short adaptive quiz and accumulated quiz performance and engagement analytics from Weeks 1–6. The student profile is updated per-KU, and subsequent video delivery is adjusted accordingly. This directly addresses the personalization drift limitation identified across the literature, where systems that assess once at the start of a course fail to track how student understanding evolves over time.

3.3 Component 3 — Script Generation Agent

For each Knowledge Unit and each student readiness level, the Script Generation Agent produces a narrated lesson script structured around the CLT-bLM four-phase model (Prepare → Initiate → Deliver → End) validated in the Adaptive Microlearning study [5]. Bloom's level governs lexical complexity and question depth: novice scripts use Remember/Understand-level language with simple analogies; proficient scripts include Analyze/Evaluate tasks and comparative framing.

The agent uses a Merged hybrid prompting strategy — combining Instruction-Tuning, Step-by-Step Reasoning, Prompt Chaining, and Persona Pattern — shown in the AI-Driven Personalized Microlearning paper [5] to achieve high lesson plan completion rates with low response latency. All scripts are reviewed by a Semantic Critique sub-agent (modeled on LASEV's three-critique system [14]) that scores generated content on three dimensions:

- **Pedagogical coherence** — does the script follow the four CLT-bLM phases in order?
- **Conceptual accuracy** — do generated claims match the source lecture transcript?
- **Learner appropriateness** — is the language and complexity calibrated to the target Bloom's level?

If any score falls below threshold, the script is automatically regenerated with refined prompts. To minimize hallucination, every generated script is grounded via Retrieval Augmented Generation (RAG) in retrieved chunks from the actual lecture transcript, ensuring that explanations are anchored in the professor's own teaching rather than the LLM's general knowledge.

3.4 Component 4 — Animation and Video Synthesis Agent

SyllaByte adopts a code-based synthesis strategy rather than pixel-based diffusion generation. This decision follows evidence that diffusion models hallucinate mathematical symbols and fail at rendering precise diagrams.

For each script segment, the agent generates Manim Python code constrained to a curated subset of the Manim Community Edition API, with template-based fallbacks for common animation types (function plots, equations, geometric proofs, transformations). To address the 38% render failure rate documented in the LLM+Manim paper [9], the agent employs the hierarchical ScopeRefine debugging strategy adapted from Code2Video [3]:

(1) **Tier 1:** Identify the erroring line, feed the error trace and offending line to the LLM, and request a targeted fix.
(2) **Tier 2:** If Tier 1 fails, regenerate the entire surrounding code block.
(3) **Tier 3:** If Tier 2 fails, regenerate the entire script segment from scratch.

A maximum retry threshold prevents infinite loops, and persistently failing segments are flagged for manual review. After successful rendering, a visual Critic agent [3] evaluates rendered preview frames using a vision-language model to detect element overlap, illegible text, and off-screen content using a grid anchor system before approving each clip for delivery.

Narration is generated using high-quality text-to-speech (ElevenLabs or Microsoft Azure Speech Studio. Final video

assembly muxes the rendered Manim animation with synthesized narration using FFmpeg, with auto-generated SRT captions for accessibility.

3.5 Logic Flow and Adaptive Feedback Loop

Data flows through SyllaByte in a structured cyclical process. First, raw lecture media and syllabus documents are ingested into the Ingestion Agent, which structures the materials into the Knowledge Unit Graph. Second, the student interacts with the Assessment Agent through the initial CAT, producing a calibrated readiness profile. Third, the Script Generation Agent ingests both the KU parameters and the student profile to produce a tailored, RAG-grounded script verified by the Semantic Critique sub-agent. Fourth, the approved script passes to the Animation Agent, which renders Manim code with ScopeRefine repair until the Visual Critic approves the final output. Fifth, the rendered video is delivered to the student along with a post-video quiz. Quiz results and engagement analytics flow back into the Assessment Agent, where they accumulate until the Week 7 reassessment trigger automatically updates the student profile and recalibrates all subsequent video generation for that student.

This closed-loop design directly addresses the personalization drift limitation that no existing system has solved: SyllaByte does not require manual instructor intervention to keep student profiles current, and adaptation continues throughout the entire semester rather than being fixed at enrollment.

4 EVALUATION

The evaluation of SyllaByte follows a two-stage approach: a pre-deployment system-level technical evaluation that validates the video generation pipeline, followed by a preregistered semester-long pilot study that measures real learning outcomes with undergraduate students.

4.1 Evaluation Plan and Methodological Justification

Stage 1 — System-Level Technical Evaluation. Before any student exposure, SyllaByte's video generation pipeline is evaluated on six automated metrics, each grounded in benchmarks established by prior work:

- **Manim Render Success Rate** — the primary functional metric from the LLM+Manim benchmark [9]. Target: ≥75%, exceeding the 62% best result reported in that paper, achieved through ScopeRefine repair and a constrained Manim API.
- **CLT-bLM Structuring Score** — semantic marker alignment across the four instructional phases, benchmarked against the 85–88% range achieved by Adaptive Microlearning [5].
- **Knowledge Unit Coverage Rate** — proportion of syllabus KUs successfully converted to a deliverable micro-video. Target: ≥90%.
- **TeachQuiz Score** — adapted from Code2Video's benchmark [3], in which a vision-language model answers knowledge questions before and after watching each generated video to measure information transfer.
- **Generation Cost and Time** — per-video token count and wall-clock generation time, reported against LASEV's [14] cost reduction benchmark and Code2Video's [3] 8.8-minute, 43,100-token baseline.
- **Expert Pedagogical Review** — three instructional designers evaluate a stratified sample of 30 generated videos using the 4-level rubric from LASEV [14] (0 = incorrect, 3 = publishable). Target: ≥85% publishable rate.

Stage 2 — Pilot Study Design. A two-condition preregistered randomized study runs across one full 14-week undergraduate STEM course:

- **Condition A — SyllaByte Personalized:** micro-videos adapted to individual student knowledge profiles, with automatic mid-semester reassessment at Week 7.
- **Condition B — Control:** unmodified lecture recordings uploaded to the LMS as currently delivered.

We target $N = 30$–40 undergraduate students randomly assigned to the two conditions.

Alternative approaches considered. We evaluated three alternatives and rejected each for specific reasons. (1) A purely automated evaluation using only TeachQuiz and expert review would be faster and cheaper, but it would not measure actual learning gains in real students. (2) A short-term cross-sectional study (e.g., a single one-hour lab session) would be feasible without IRB-heavy approval, but it cannot detect personalization drift, retention decay, or transfer effects that only emerge over weeks. (3) A larger three-arm RCT with a non-personalized SyllaByte arm would more cleanly isolate the effect of personalization, but recruiting and managing 180 students at NYUAD over a full semester is logistically infeasible within a two-semester capstone. The chosen two-condition pilot is the most rigorous design that is realistically achievable.

4.2 Experimental Setup and Outcome Measures

The pilot study collects data at three time points: baseline (Week 0), midterm (Week 7), and end of semester (Week 14).

Primary outcome measures.

- **Knowledge Retention** — validated instructor-designed quizzes aligned to each Knowledge Unit, administered immediately after each video module and again two weeks later, enabling within-semester forgetting curve analysis.
- **Transfer Performance** — novel application-level questions on the final exam (Bloom's Apply/Analyze tier) not directly covered in any video.
- **Course Grade** — final course grade as an ecologically valid measure of academic performance.
- **Engagement Metrics** — LMS-logged watch time, replay frequency, drop-off timestamps, and in-video quiz response rates, captured using xAPI behavioral analytics [1].

Secondary outcome measures.

- **Cognitive Load** — NASA-TLX administered after each module.
- **System Usability** — System Usability Scale (SUS), benchmarked against the 82.1 score achieved by Adaptive Microlearning [5].
- **Student-Perceived Learning Efficacy** — a custom 7-point Likert questionnaire covering comprehension, engagement, relevance, and pacing, modeled on ReelEd's questionnaire [10].
- **Trust in AI-Generated Content** — Jian et al.'s 16-item trust in automation scale, adapted by ReelEd [10], measuring both competence-based and benevolence-based trust.

Qualitative component. At Week 7 and Week 14, semi-structured focus groups (n=6 students per condition) capture perceived accuracy and trustworthiness of AI-generated content, preferences for control and transparency, and unexpected use patterns.

Analysis plan. The primary analysis uses Linear Mixed-Effects Models (LMM) with repeated measures (Week 0, 7, 14), condition (A/B) as fixed effect, and student as random effect, accommodating within-semester learning trajectories. Effect sizes (Cohen's d) and 95% confidence intervals will be reported for all primary outcomes. Pre-registration will be filed on OSF prior to data collection. Secondary analyses will use ANCOVA controlling for baseline knowledge scores, and qualitative data will be analyzed using Reflexive Thematic Analysis.

4.3 Hypotheses and Anticipated Results

The evaluation tests four primary hypotheses, each grounded in findings from the reviewed literature:

- **H1 (Retention):** Students in Condition A will achieve significantly higher knowledge retention scores than those in Condition B. We anticipate an effect size of approximately d=0.5, in line with the 15-percentage-point retention gain reported in the Adaptive Microlearning study [5].
- **H2 (Engagement):** Students in Condition A will exhibit higher watch-completion rates and lower drop-off rates than those in Condition B. We anticipate completion rates approaching the 87% reported by Adaptive Microlearning [5] versus the 52% baseline for full-length videos.
- **H3 (Personalization Value):** Students will show measurable learning gains following the automatic Week 7 reassessment, demonstrating that adaptive recalibration mid-semester improves outcomes over a static initial profile. This addresses the personalization drift gap identified in the GenAI Pedagogy Framework [1].
- **H4 (Usability):** SyllaByte will achieve a System Usability Scale score $\geq$80, comparable to the 82.1 score reported in Adaptive Microlearning [5], indicating that the automated pipeline produces content that students find usable and acceptable.

Preliminary results. As this is a proposal, no full pilot data is available. However, preliminary technical evaluation of an early Manim generation prototype indicates that constrained-API generation combined with single-tier ScopeRefine repair achieves render success rates in the 65–70% range. This suggests that the $\geq$75% target with full three-tier ScopeRefine is realistic. Initial CLT-bLM structuring scores on a sample of 20 generated scripts ranged from 82–87%, in line with the Adaptive Microlearning benchmark.

4.4 Visualizations of Anticipated Results

The final evaluation will include three primary visualizations to communicate findings:

- **Forgetting curve comparison** — a line plot showing retention scores over time (immediate post-test, 1-week, 2-week, and end-of-semester recall) for both conditions, modeled on the visualization style used in the Adaptive Microlearning paper [5].
- **Pre/post personalization profile shifts** — a heatmap showing each student's per-KU readiness level at Week 0 and after the Week 7 reassessment, demonstrating that the automatic recalibration mechanism captures real changes in student understanding.
- **Engagement timeline** — a stacked area chart of weekly watch-completion rates per condition, showing whether the SyllaByte personalized condition sustains engagement throughout the semester or whether engagement decays in line with traditional video formats.

Failure mode considerations. The evaluation explicitly tests for three failure modes identified in the reviewed literature: (1) AI hallucination, addressed by an automated factual consistency check using embedding similarity against the lecture transcript; (2) personalization drift, addressed by the automatic Week 7 reassessment; and (3) social presence deficit, mitigated by optionally including the professor's voice and a brief on-camera introduction to each video module. Ethical safeguards include IRB approval submitted at the start of Semester 1, opt-out provisions ensuring no student is disadvantaged by declining participation, FERPA-compliant data storage with no personally identifiable information in LLM prompts, and explicit student disclosure that content is AI-generated.

5 PROJECT TIMELINE

The SyllaByte project is structured across two semesters: Fall 2026 (September–December) for system development and technical evaluation, and Spring 2027 (January–April) for the pilot study, data analysis, and final write-up. IRB approval is pursued in parallel with development to ensure it is in place before the pilot study begins.

5.1 Fall 2026 — System Development and Technical Evaluation

September — Foundations and IRB Submission.

- Set up monorepo with separate packages for each agent.
- Configure Docker containers for reproducible Manim and Whisper environments.
- Set up MongoDB, object storage, and a Celery + Redis job queue for asynchronous video generation.
- Submit IRB application for the Spring 2027 pilot study.
- Identify and confirm a cooperating STEM professor and course for the pilot.
- Build syllabus parser using PyMuPDF and python-docx with LLM-based structured extraction.

October — Components 1 and 2: Ingestion and Knowledge Assessment.

- Implement lecture audio transcription pipeline using OpenAI Whisper with timestamp alignment.
- Build the Knowledge Unit Graph using semantic chunking, LLM-based concept extraction, Bloom's taxonomy tagging, and NetworkX for graph structure.
- Generate per-KU item banks using LLM prompting at varying Bloom's difficulty levels and manually validate a subset.
- Implement CAT engine using the catsim library with the 2PL IRT model.
- Build the scheduled job system for the automatic mid-semester reassessment.

November — Component 3: Script Generation.

- Design CLT-bLM prompt templates with placeholders for all four phases.
- Implement merged hybrid prompting strategy combining instruction tuning, chain of thought, prompt chaining, and persona pattern.
- Build the Semantic Critique sub-agent for pedagogical coherence, conceptual accuracy, and learner appropriateness scoring.
- Implement RAG grounding using Sentence-BERT embeddings against lecture transcripts to mitigate hallucination.

December — Component 4: Animation, Integration, and Technical Evaluation.

- Build Manim code generation pipeline with few-shot prompting and a curated API subset.
- Develop template-based fallbacks for common animation types (function plots, equations, geometric proofs).
- Implement the three-tier ScopeRefine debugging loop with retry threshold.
- Build the Visual Critic agent using a vision-language model for layout and legibility verification.
- Integrate ElevenLabs or Azure TTS for narration and FFmpeg for final video assembly with SRT captions.
- Integrate all four components into a unified end-to-end pipeline.
- Build the student-facing web interface (Next.js) and quiz delivery system.
- Conduct full Stage 1 technical evaluation: Manim render success rate, CLT-bLM structuring score, KU coverage rate, TeachQuiz score, generation cost and time, and expert pedagogical review of 30 sample videos.
- Confirm IRB approval status and finalize student recruitment plan.

5.2 Spring 2027 — Pilot Study, Analysis, and Write-Up

January — Pilot Study Launch (Weeks 0–3).

- Recruit 30–40 STEM undergraduate students after IRB approval is confirmed.
- Conduct system onboarding and obtain informed consent.
- Administer Week 0 baseline: CAT knowledge assessment, demographic survey, and baseline knowledge quiz.
- Begin Module 1 video delivery for Condition A (SyllaByte Personalized) and standard lecture recordings for Condition B (Control).

- Begin continuous collection of LMS engagement analytics and post-video quiz responses.

February — Module 1 Completion and Mid-Semester Reassessment (Weeks 4–7).

- Continue Module 1 delivery and data collection.
- Trigger automatic mid-semester profile reassessment at Week 7.
- Administer the midterm battery: retention quiz, NASA-TLX, SUS, Likert survey on perceived learning efficacy, and trust in automation scale.
- Conduct first round of semi-structured focus groups (n=6 students per condition).
- Begin Module 2 delivery using updated personalized profiles for Condition A.

March — Module 2 Delivery and Final Assessment (Weeks 8–14).

- Continue Module 2 delivery and data collection.
- Administer two-week delayed recall quizzes for Module 1 content to enable forgetting curve analysis.
- Monitor system performance and address any technical issues that arise during deployment.
- Administer final exam including transfer-tier questions on novel application content.
- Administer the full final battery of scales: NASA-TLX, SUS, Likert questionnaire, and trust scale.
- Conduct end-of-semester focus groups with students.
- Collect final course grades for ecologically valid outcome measurement.

April — Data Analysis and Final Write-Up.

- Run primary analysis using Linear Mixed-Effects Models on retention, transfer, course grade, and engagement outcomes.
- Run secondary ANCOVA analyses controlling for baseline knowledge.
- Conduct Reflexive Thematic Analysis on focus group transcripts.
- Generate visualizations: forgetting curve comparison, pre/post personalization profile shifts, and engagement timeline.
- Complete final capstone report integrating system design, technical evaluation results, pilot study findings, and qualitative insights.
- Prepare and deliver final capstone presentation.
- Submit pre-registration documentation to OSF and prepare supplementary materials including the open-source code repository.

5.3 Risk Management and Contingency Planning

The timeline includes flexibility for the highest-risk activities. If Manim render success rates fall below 75% in December, up to two weeks of additional debugging are available before the January pilot launch. If IRB approval is delayed, a fallback synthetic evaluation using LLM judges and a smaller volunteer pilot of 5–10 students will be initiated. If recruitment falls short of 30 students, the analysis plan will adjust effect size targets and report results as a preliminary pilot.

6 BUDGET

The proposed budget covers compute resources, API access for LLM-based components, software licenses, and participant compensation for the pilot study. All requests are in USD and reflect realistic estimates based on current pricing as of 2026. The budget has been scoped to minimize cost by leveraging free-tier resources, student/research API credits, NYUAD-provided infrastructure, and local compute wherever possible.

6.1 LLM API Costs

Anthropic Claude API (script generation, critique, code generation): $300.

The Script Generation Agent and Semantic Critique subagent rely on Claude Opus and Claude Sonnet for high-quality script generation, critique, and Manim code generation.

OpenAI API (GPT-4o for visual critique and fallback): $150.

The Visual Critic agent uses GPT-4V (or equivalent) for vision-based layout verification of rendered Manim frames.

ElevenLabs or Azure TTS (narration generation): $50.

High-quality narration is produced using ElevenLabs or Microsoft Azure Speech Studio.

6.2 Compute Infrastructure

Local rendering and minimal cloud hosting: $100.

Manim rendering, the Celery + Redis job queue, and the FastAPI backend will run on a personal development machine during development rather than a dedicated cloud VM. A small cloud instance (~$15/month) is budgeted only for the pilot study deployment period to host the student-facing web interface reliably.

Object storage for video files and assets: $0.

Generated videos, lecture recordings, and intermediate artifacts will be stored on AWS S3 free tier (5GB) and NYUAD-provided storage, which is sufficient for the pilot scale.

6.3 Pilot Study Costs

Participant compensation: $600.

Compensating 30 undergraduate participants with $20 gift cards each for completing the full semester-long pilot, including baseline assessment, midterm battery, focus groups, and end-of-semester assessment. Compensation is structured as completion-based to maintain participation across the 14-week study.

Focus group refreshments and materials: $50.

Light refreshments for the four focus group sessions across the semester (two midterm sessions and two end-of-semester sessions).

Expert pedagogical reviewer honoraria: $50.

Three instructional design experts will evaluate a stratified sample of 30 generated videos using the LASEV 4-level rubric. Reviewers will be recruited from NYUAD faculty and graduate students on a volunteer basis, with a compensation ($15–$20 each).

6.4 Software and Tools

Open-source primary stack: $0.

The core development stack — Python, Manim Community Edition, Whisper, FastAPI, Celery, Redis, MongoDB, NetworkX, Sentence-BERT, catsim, MoviePy, FFmpeg — is fully open source and free to use.

Optional development tools: $50.

This is a small allowance for any paid subscriptions needed during development.

6.5 Budget Summary

- LLM API costs (Anthropic, OpenAI, TTS): $500
- Compute infrastructure: $100
- Pilot study (compensation, materials, expert reviewers): $700
- Software and tools: $50
- Contingency buffer (~10%): $135

Total requested budget: $1,485 USD.

7 CONCLUSION

This proposal has presented SyllaByte, a multi-agent AI pipeline that ingests a professor's syllabus and lecture recordings to automatically generate personalized 2–5 minute animated micro-videos for university STEM education. The system effectively addresses four persistent gaps in the literature: course-grounded generation, automated mid-semester personalization, reliable Manim synthesis, and longitudinal evaluation. If successful, SyllaByte will deliver the first syllabus-grounded AI video generation system and the first semester-long controlled pilot study in a real STEM course.

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